Hari Bhavsar

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Research Interests

I am broadly interested in Computer Architecture and Memory Systems. My research focuses on secure caches as well as domain specific cache and memory system design.

Work Experience

Member, Tech *June 2026 – Present*
D.E. Shaw India

Education

Indian Institute of Technology Bombay *October 2022 – May 2026*
B.Tech. in Computer Science and Engineering (with Honors) GPA: 9.33/10

Publications

- A. Bhatla†, H. Bhavsar†, S. Saha, B. Panda. "So, You Think You Know All About Secure Randomized Caches?" *USENIX Security (2025).* Paper, Talk, Artifact. Distinguished Artifact Award

Research Experience

SoK: So, You Think You Know All About Secure Randomized Caches? *May 2024 – August 2025*
Prof. Biswabandan Panda & Prof. Sayandeep Saha, IIT Bombay

- Investigated **conflict based** cache side channel attacks that exploit the **set associativity** of Last Level Caches to obtain hidden information from victim processes and studied designs to mitigate these attacks
- Quantified LLC security by computing the size and probability of generating useful **eviction sets**
- Classified various **microarchitectural modifications** (knobs) and analyzed their individual and collective effects on LLC security, concluding that **high associativity** is the most crucial knob for providing security
- Evaluated the security of different knobs against **cache occupancy based** side channel attacks, concluding only **cache partitioning** provides sufficient security against them but at the cost of performance

GALE: Graph-Aware Exclusive Last-Level Cache *December 2024 – December 2025*
Prof. Biswabandan Panda, IIT Bombay

- Extended the **ChampSim** simulator to model an exclusive last level cache (LLC), finding that an exclusive LLC provides **speedup of 11.7%** as compared to a non-inclusive LLC
- Used **valgrind** to profile graph-based workloads and found that the 20% of vertices having **the most edges** account for 80% of Last Level Cache accesses, indicating that vertex degree could be a replacement heuristic
- Discovered that cache blocks containing hot vertices obtain more **level two cache hits** and used this to build an LLC bypass policy, obtaining a speedup of 21% at a storage overhead of **2KB** at the L2

Uncovering when and why off-chip predictors work *December 2025 – May 2026*
Under submission to TACO
Prof. Biswabandan Panda, IIT Bombay

- Showed that off-chip predictors (OCPs) such as hermes, even when accurate in prediction, can **increase DRAM congestion** and **degrade performance** by prematurely discarding correctly predicted requests
- Discovered that the performance improvement caused by OCPs is highly dependent on the available DRAM bandwidth and that **large improvements require low DRAM contention**
- Identified workloads in which OCPs **increase the stall cycles** at the re-order buffer and thus degrade performance

The Final Exclusive Last-Level Cache Replacement Policy *July 2026 – Present*

- Evaluated state of the art last-level cache (LLC) replacement policies on an exclusive LLC and found that they all **degrade performance** compared to LRU despite providing speedup on a non-inclusive LLC
- Discovered that **IP-based** replacement policies such as Mockingjay are ineffective on datacenter workloads as those workloads have over 1000x the **instruction footprint** as compared to other workloads
- Implemented an **IP and criticality** based replacement policy for an exclusive LLC to provide consistent speedup

Honours & Awards

- Distinguished Artifact Award, USENIX Security (2025)
- Excellence in Teaching Assistantship (2025)
- Achieved an **AP** grade in Advanced Computer Architecture, given to the top 2 out of 105 students (2024)

Internships

Tracking Dependencies for Memoized Functions *May 2025 – July 2025*
D.E. Shaw India

- Generated and stored the dependencies of **disk** and **RAM memoized** Python functions using Python's **coverage** module for code-level dependencies and Linux's **strace** tool for file-level dependencies
- Identified the files directly modified by a **git commit** and generated **abstract syntax trees** for Python files to extract the specific code classes and functions that had been affected by the commit

Teaching & Mentorship Experience

Project Mentor *May 2024 – August 2024*
Seasons Of Code, IIT Bombay

- Taught a group of **22** people about the fundamentals of **parallel programming**, such as **SIMD** instructions and **multi-threading** as well as how one could exploit these techniques for performance gains
- Released content **weekly** and kept meetings to ensure that mentees were able to progress consistently
- Guided mentees through learning the **Open Computing Language** in order to code on the **GPU**
- Provided mentees with naive implementations of a **SAT Solver** and a **Machine Learning** library, which they were able to significantly optimize using the aforementioned techniques to get large performance gains

Teaching Assistant *January 2025 – May 2025*
Operating Systems Lab, IIT Bombay

- Created **lab exams** and conducted weekly labs for a group of over 180 students
- Implemented **process threads**, and concurrency primitives such as **locks**, **barriers**, **conditional variables** and **semaphores** in Xv6 and gave detailed instructions to students on how to do the same

Teaching Assistant *August 2025 – December 2025*
Digital Logic Design & Computer Architecture, IIT Bombay

- Conducted **weekly labs**, created **assignments, slides** and **videos** to enhance student learning
- Structured the course by dividing it into four core components, hardware design in **Verilog**, **x86 assembly**, **performance analysis** and understanding microarchitecture through the **ChampSim simulator**

Selected Academic Projects

Hypervisor Design and Linux Kernel Modules *Jan 2025 – May 2025*
Prof. Purushottam Kulkarni

- Used the **IOCTL** syscall to create a **device driver** that provided interfaces for reassigning child and parent processes, given process IDs, as well as one to terminate all child processes of a given process
- Utilized the Linux KVM to create a **Paravirtualized Guest Operating System** with separate modes for memory management, and with **hypercalls** to perform memory translations and file IO handling
- Implemented a method to keep track of process stats using the **procfs** and **sysfs** Linux Kernel Modules

Compiler Frontend Design *January 2025 – May 2025*
Prof. Uday Khedker

- Implemented a **scanner** and a **parser** for a simplified version of C containing arithmetic operations, function calls and conditional statements, using an LALR(1) parser from **bison** and **flex** for scanning
- Conducted **semantic analysis** of the language by constructing an **abstract syntax tree** for type-checking and generating **three address code** for the language, ensuring the language was strongly typed
- Ensured support for **global variables** by using a multi-layered symbol table for book-keeping of variables

Image Compression Framework *August 2024 - December 2024*
Prof. Ajit Rajwade

- Implemented a lossy image compression framework for grayscale images by computing the **Discrete Cosine Transforms** of the quantized images, then using Huffman Encoding and RLE for lossless compression
- Incorporated downsampling in the chrominance components of colored images and provided APIs to specify the quality factor from 0 to 255, achieving a compression ratio of **6:1** at a quality factor of 10

Technical Skills

Languages	C, C++, Python, x86 assembly, MIPS, bash, sed, awk, javascript, OpenCL
Tools	Git, Linux, Docker
Simulators	ChampSim, Valgrind, CacheFX, cache-model
Areas	Computer Architecture, Operating Systems, Compilers